4. Line L_1 passes through the points $\mathrm{A}(-3, 0.5)$ and $\mathrm{B}(9, -3.5)$.

(a) Find the gradient of L_1. [2]

Line L_2 passes through the point $\mathrm{C}(3, 1)$ and is parallel to L_1.

(b) Determine the equation of L_2. Give your answer in the form $ax + by + d = 0$, where a, b and d are integers. [2]

(c) Find the coordinates of the x-intercept of L_2. [2]

Working:

Calculator allowed

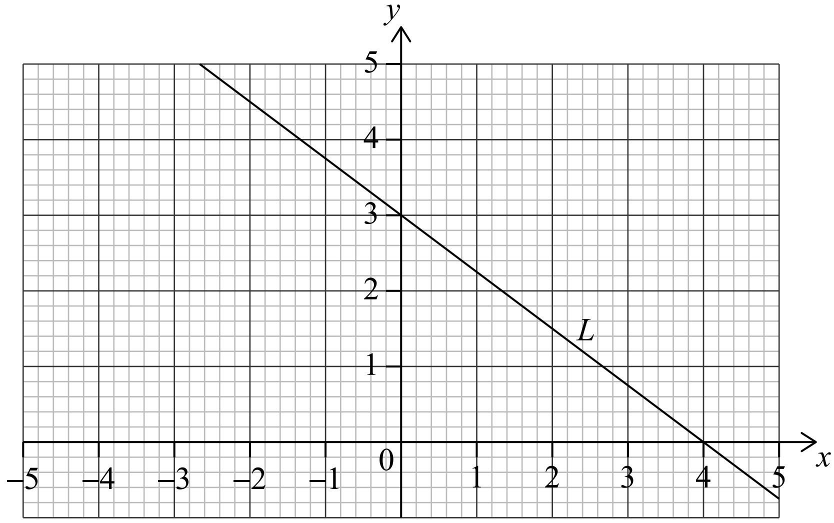
Answers:

(a) . . . .

(b) . . . .

(c) . . . .

8. Line L has a y-intercept at $(0, 3)$ and an x-intercept at $(4, 0)$, as shown on the following diagram.



(a) (i) Find the gradient of L.

(ii) Write down the equation of L in the form $y = mx + c$. [3]

Line N is perpendicular to L, and passes through point $\mathrm{P}(2, 1)$.

(b) (i) Write down the gradient of N.

(ii) Find the equation of N in the form $y = mx + c$. [3]

Working:

Answers:

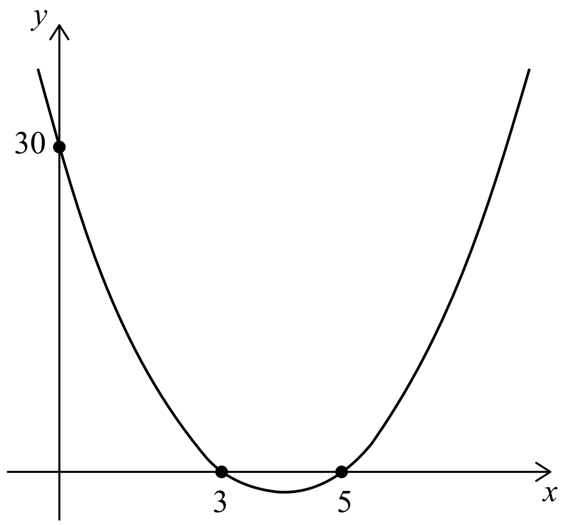
(a) (i) . . . .

(ii) . . . .

(b) (i) . . . .

(ii) . . . .

13. The graph of a quadratic function is shown.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + 30$. [4]

(b) Write down the equation of the axis of symmetry. [2]

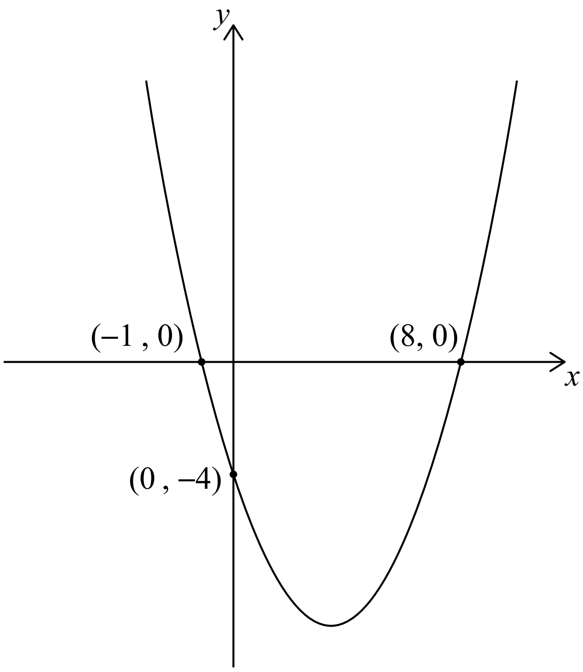
Working:

Answers:

(a) . . . .

(b) . . . .

10. Consider the function $f(x) = a(x - p)(x - q)$, shown on the following graph.



(a) Find the equation of the axis of symmetry. [2]

(b) Find the value of p and of q. [2]

(c) Find the value of a. [2]

Working:

Answers:

(a) . . . .

(b) . . . .

(c) . . . .

3. [Maximum mark: 7]

Let $g(x) = x^2 + bx + 11$. The point $(-1, 8)$ lies on the graph of g.

(a) Find the value of b. [3]

(b) The graph of $f(x) = x^2$ is transformed to obtain the graph of g.

Describe this transformation. [4]

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5. [Maximum mark: 6]

Consider the function $f(x) = (1 - k)x^2 + x + k$, $x \in \mathbb{R}$. Find the value of k for which $f(x)$ has two equal real roots.

2. [Maximum mark: 6]

Consider the graph of the function $f(x) = a(x + 10)^2 + 15$, $x \in \mathbb{R}$.

(a) Write down the coordinates of the vertex. [2]

(b) The graph of f has a y-intercept at -20. Find a. [2]

(c) Point $\mathrm{P}(8, b)$ lies on the graph of f. Find b. [2]

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. In particular, solutions found from a graphic display calculator should be supported by suitable working, for example if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines if necessary.

1. [Maximum mark: 6]

A group of 7 adult men wanted to see if there was a relationship between their Body Mass Index (BMI) and their waist size. Their waist sizes, in centimetres, were recorded and their BMI calculated. The following table shows the results.

Waist (x cm)	58	63	75	82	93	98	105
BMI (y)	19	20	22	23	25	24	26

The relationship between x and y can be modelled by the regression equation $y = ax + b$.

(a) (i) Write down the value of a and of b.

(ii) Find the correlation coefficient. [4]

(b) Use the regression equation to estimate the BMI of an adult man whose waist size is 95 cm. [2]

. . . .

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. In particular, solutions found from a graphic display calculator should be supported by suitable working, for example if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines if necessary.

1. [Maximum mark: 6]

The number of messages, M, that six randomly selected teenagers sent during the month of October is shown in the following table. The table also shows the time, T, that they spent talking on their phone during the same month.

Time spent talking on their phone (T minutes)	50	55	105	128	155	200
Number of messages (M)	358	340	740	731	800	992

The relationship between the variables can be modelled by the regression equation $M = aT + b$.

(a) Write down the value of a and of b. [3]

(b) Use your regression equation to predict the number of messages sent by a teenager that spent 154 minutes talking on their phone in October. [3]

14. Devra invested k US dollars (USD) in an account that pays a nominal annual interest rate of 3.1%, **compounded monthly**. After 6 years she has 1100 USD in the account.

(a) Calculate the value of k. **Give your answer to 2 decimal places.** [3]

Devra then bought a computer that cost 1100 USD and sold it 4 years later for 350 USD.

(b) Find the rate at which the computer depreciated per year. [3]

Working:

Answers:

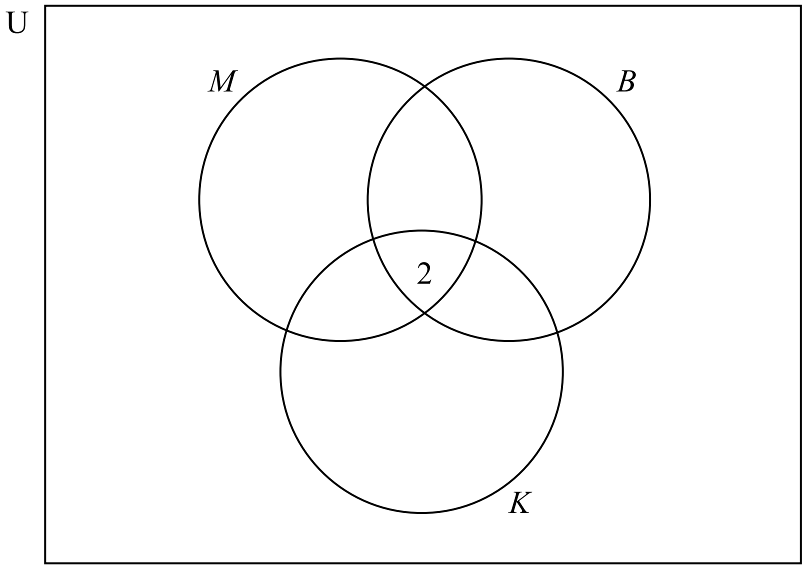
(a) . . . .

(b) . . . .

5. A school café sells three flavours of smoothies: mango (M), kiwi fruit (K) and banana (B). 85 students were surveyed about which of these three flavours they like.

35 students liked mango, 37 liked banana, and 26 liked kiwi fruit
2 liked all three flavours
20 liked both mango and banana
14 liked mango and kiwi fruit
3 liked banana and kiwi fruit

(a) Using the given information, complete the following Venn diagram. [2]



(b) Find the number of surveyed students who did not like any of the three flavours. [2]

(c) A student is chosen at random from the surveyed students.

Find the probability that this student likes kiwi fruit smoothies given that they like mango smoothies. [2]

Working:

Answers:

(b) . . . .

(c) . . . .

4. [Maximum mark: 6]

Let $f(x) = \dfrac{2x-1}{x+3}$, $x \neq -3$.

(a) Write down the equation of the vertical asymptote of the graph of f. [1]

(b) Find $f^{-1}(x)$. [3]

(c) Find the equation of the horizontal asymptote of the graph of f^{-1}. [2]